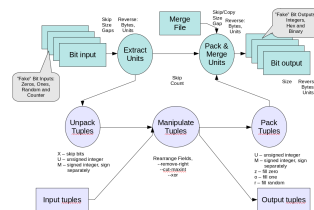


## bdd — Bit Data Dump

High-performance CLI tool and Rust library to interpret, manipulate, stream, merge, and pack arbitrary-width bitstreams.

Originally created by Esa Turtiainen in Python 2 (2010), 2026 `bdd` was re-engineered in modern Rust for memory safety, improved maintainability, and multi-gigabit throughput with zero external C dependencies.

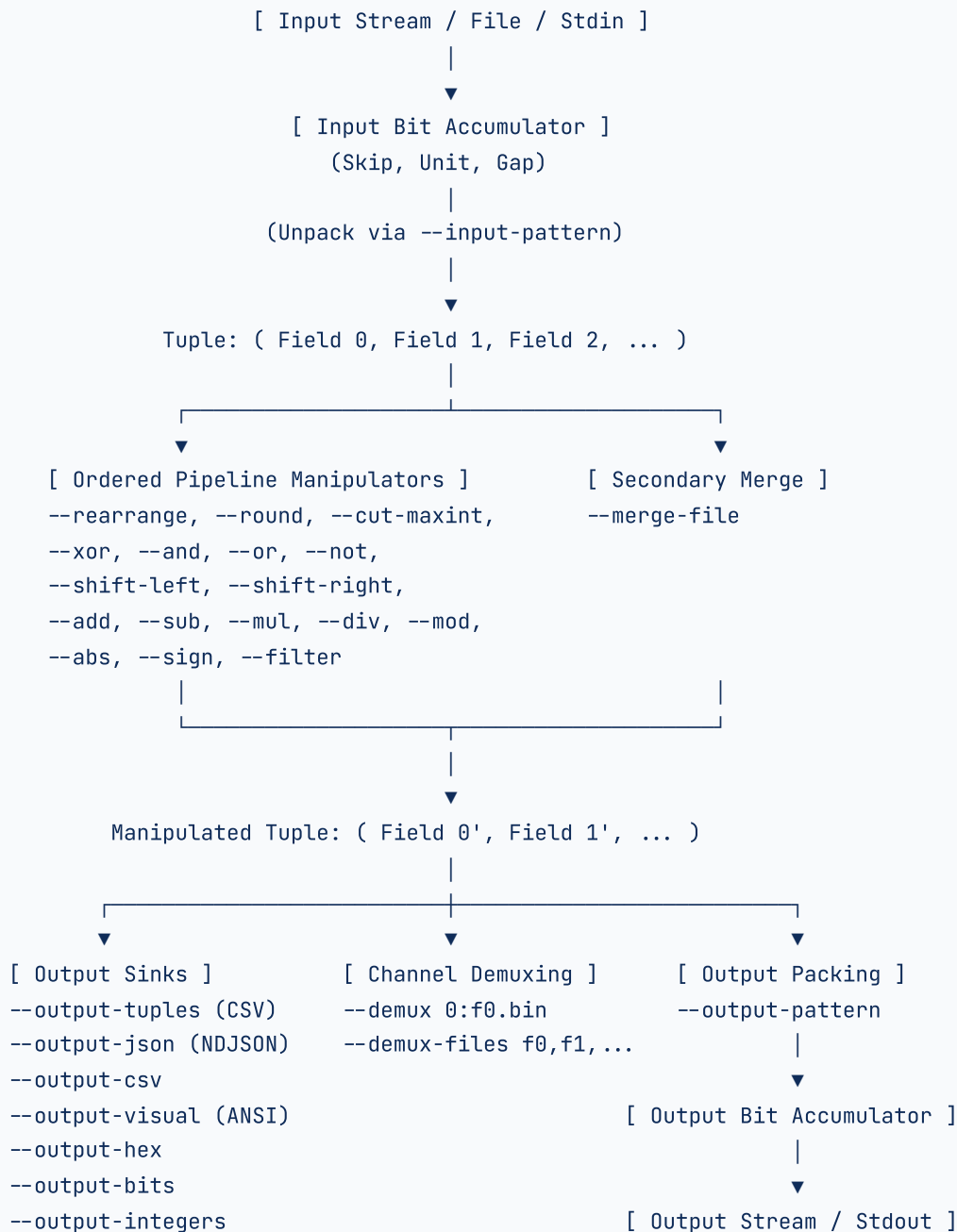
bdd is available as Debian and RPM packages, Rust library, Python library and C library.



## What is bdd?

Most Unix tools ( `dd` , `hexdump` , `od` , standard shell pipes) operate strictly on byte (8-bit) or machine-word (16, 32, 64-bit) boundaries. Real-world binary streams—such as embedded sensor telemetry, cryptographic protocols, SDR/radio frames, network headers, compressed payloads, and legacy archives—frequently pack unaligned fields of arbitrary bit lengths (e.g., 1-bit flags, 3-bit opcodes, 12-bit ADC samples, or 57-bit mantissas).

`bdd` is a **binary-dd for bitstreams**: it allows you to slice, unpack, inspect, transform, interleave, and repack streams of any bit width with single-bit precision.



## 1. Anatomy of a Bit Stream: Units, Skips, and Gaps

### The Default Baseline: 8-Bit Streaming

The simplest invocation of `bdd` establishes its fundamental operating principle:

```
bdd < a > b
```

This command copies file `a` to file `b` in **default 8-bit units** (1 byte at a time) from standard input to standard output, without any modifications (acting like a byte-exact `cat` or `dd`).

By default:

- **Default Input Unit:** 8 bits (1 byte)
- **Default Output Unit:** 8 bits (1 byte)
- **Default Gaps & Skips:** 0 bits

Every slicing and packing feature in `bdd` builds on top of this baseline. For example, if you change `--input-unit=3` without setting an output unit:

```
# Create a stream of dense 3-bit units (top 5 bits of each byte are ignored):
bdd --output-unit=3 < foo.8bit > foo.3bit

# Extract 3-bit stream and expand each into an 8-bit byte with 5 leading zero bits:
bdd --input-unit=3 < foo.3bit > foo.8bit

# (If the top 5 bits were zeros in foo.8bit, original and round-tripped foo.8bit are identical)
```

Because the output unit remains at its default of 8 bits, each 3-bit input unit ( `xxx` ) is padded on the left with five zero bits ( `00000xxx` ) to produce an 8-bit byte on standard output.

### **WARNING**

**Sub-Byte Output Packing & End-of-Stream Zero Padding** Unix files, pipes, and block devices operate strictly on 8-bit bytes. When packing units smaller than 8 bits (e.g. `--output-unit=1` or `--output-unit=3` ) whose total count does not sum to an exact multiple of 8 bits, the final byte flushed to output is padded with trailing zero bits to reach a byte boundary. That means that if the last byte contains, for example, **three 1-bit units** ( `1, 0, 1` ), `bdd` flushes a full 8-bit byte containing those three bits followed by five synthetic zeros ( `10100000` = `0xA0` ). When that file or pipe is subsequently read as bits (e.g. `--input-unit=1` ), the reader will receive **eight bits** (the 3 original units plus 5 extra zero bits). **The count of what you write is not necessarily the count of what you read.** If exact unit preservation is important, you have to count units yourself:

- Specify `--count=N` when reading the stream back (e.g. `bdd --input-unit=1 --count=3 < stream.bin` ), or
- Frame sub-byte units within explicit byte-aligned containers (e.g. `8[0:1]` ), or
- Store the exact unit count or bit length in external metadata or an application header.

## Units, Skips, and Gaps (The Simple Model)

In `bdd` , all stream dimensions are measured strictly in **bits**, not bytes. In the basic stream model, data is processed as a repeating sequence of **Units**:

Stream Start



- `--input-unit=BITS` ( `-u` ): The number of bits in each repeating processing unit (default: `8` ).
- `--input-skip-bits=BITS` : Initial offset in bits skipped before processing the first unit (default: `0` ).
- `--input-skip-units=UNITS` : Skip initial N units from stream start.
- `--input-gap=BITS` : Number of bits skipped *after* reading each unit (default: `0` ).
- `--input-assert-aligned` : Aborts with an error if the input stream terminates unaligned to a byte boundary.

## Basic Slicing Recipes

- **Extract the Most Significant Bit of every byte** (expands 1 bit to an 8-bit byte `0x00` or `0x01` ):

```
bdd --input-unit=1 --input-gap=7 --output-unit=8 < input.bin
```

- **Pack the Least Significant Bit of bytes into a dense bitstream** (e.g. 9 bytes with LSB=1 become two bytes `0xFF` and `0x80` ):

```
bdd --input-skip-bits=7 --input-unit=1 --input-gap=7 --output-unit=1 < input.bin
```

## The Next Challenge: Physical Containers in Real-World Data

The simple unit model works when an entire file is a uniform sequence of identical units. But in real-world binary formats (telemetry, network packets, audio words, video frames), active fields don't float freely in an empty void—they sit inside fixed physical containers: bytes (8 bits), half-words (16 bits), words (32 bits), audio stereo frames (32 bits), or MPEG-TS packets (188 bytes).

If you want bits 3 and 4 of every byte, the simple unit model forces you to calculate an initial skip of 2 bits, an active unit of 2 bits, and a trailing gap of 4 bits to reach the next byte. Worse: if you later need bit 5 instead, you must manually recalculate both the initial skip (4 bits) and the trailing gap (3 bits).

**What is missing?** A way to declare the fixed outer container size once, and simply point to the active field inside it. This is solved in Chapter 2.

## 2. Containers & Raw Units: Solving the Offset Problem

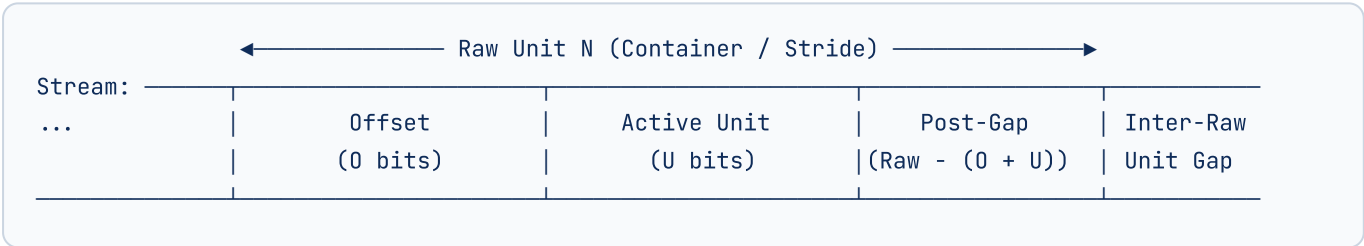
In structured binary files, active fields are rarely isolated in a continuous stream; instead, they reside inside fixed-size physical containers: bytes (8 bits), half-words (16 bits), words (32 bits), audio stereo frames (32 bits), or network headers.

### How Raw Units Differ from Units

- A **Unit** ( `--input-unit` ) is the *active payload* you extract and process (e.g. 2 bits).
- A **Raw Unit** ( `--input-raw-unit` ) is the *outer container, frame, or stride* (e.g. 8 bits or 32 bits) in which the active unit lives.

**The mental arithmetic problem with simple units:** If you want bits 3 and 4 of every byte, simple unit mode requires calculating an initial skip of 2 bits, an active unit of 2 bits, and a trailing gap of 4 bits to reach the next byte. If you subsequently need bit 5, you have to recalculate both the initial skip (4 bits) and the trailing gap (3 bits).

**The solution with raw units:** Declare the container size once with `--input-raw-unit=8` . Then simply state the field's starting position with `--input-offset` and field width with `--input-unit` . `bdd` automatically derives the initial skip and skips the trailing remainder of the container ( $R - (O + U)$ ).



- `--input-raw-unit=BITS` : The repeating container size in bits.
- `--input-offset=BITS` : Bit offset of the active unit within each raw unit (default: `0` ). Requires `--input-raw-unit` .
- `--input-unit=BITS` : Width of the active unit (defaults to `raw-unit - offset` ).
- `--input-gap=BITS` : When `--input-raw-unit` is set, defines the gap *between* successive raw units (default: `0` ).

## Clean Slicing Recipes with Raw Units

```
# Extract bits 3 and 4 of every byte (offset 2, length 2, trailing 4 bits skipped
automatically):
bdd --input-raw-unit=8 --input-offset=2 --input-unit=2 --output-integers < input.bin

# Extract bit 5 of every byte (offset 4, length 1, trailing 3 bits skipped automatically):
bdd --input-raw-unit=8 --input-offset=4 --input-unit=1 --output-integers < input.bin

# Extract 16-bit audio channel from interleaved 32-bit stereo (L=0..15, R=16..31):
bdd --input-raw-unit=32 --input-offset=0 --input-unit=16 < audio.raw # Left channel
bdd --input-raw-unit=32 --input-offset=16 --input-unit=16 < audio.raw # Right channel
```

### The Next Challenge: Command-Line Verbosity & Symmetrical Framing

Declaring containers with `--input-raw-unit` and `--input-offset` eliminates mental arithmetic. But typing out `--input-raw-unit=8 --input-offset=2 --input-unit=4 --input-gap=8 --output-raw-unit=8 --output-offset=2 ...` is verbose, repetitive, and cumbersome to type interactively. Furthermore, while incoming containers are sliced cleanly, framing the *output* stream into structured containers required configuring four separate output flags.

**What is missing?** A visual, concise shorthand notation that expresses the entire input container slicing and output framing transformation in a single, intuitive argument. This is solved in Chapter 3.

## 3. Unified Stream I/O Mapping: Solving Command-Line Verbosity

Instead of specifying separate flags for skips, containers, offsets, and gaps, `bdd` accepts a unified, expressive **stream I/O mapping expression** as an optional positional argument:

```
bdd "<input_stream> → <output_stream>" [input_pattern] [output_pattern] [options]
```

Each stream definition uses the syntax:

```
[skip :] unit_or_container [+ gap]
```

- **Initial Skip / Prefix ( : )**: Delimited by `:`. Skips  $N$  bits on input, or emits  $N$  zero bits as an initial stream prefix on output.
- **Periodic Gap ( + )**: Appended with `+`. Defines bits skipped between input containers, or zero bits emitted between output containers.
- **Containers & Bitfields**: Can be expressed in two complementary formats:
  - **Form A (Container Slicing)**: `raw_size[offset : unit]` or `raw_size[offset, unit]` — container size outside, offset and active unit inside (post-gap is computed automatically as

raw — offset — unit). Slice components can be separated by either `:` or `,`.

- **Form B (Physical Layout Box):** `[pre : unit : post]` or `[pre, unit, post]` — explicit pre-gap, active payload, and post-gap inside the box (container size is the sum `pre + unit + post`).
- **Bare Unit:** A plain number or size (e.g. `8`, `3`, `188B`, `4k`).

## Multiplications, Suffixes & Large Units ( `k`, `M`, `G` )

Every numeric size in a stream pattern (skips, container sizes, offsets, units, gaps) fully supports standard binary/decimal suffixes and multiplication expressions:

- **Binary Suffixes (powers of 1024):** `k` / `KiB` ( $1,024$ ), `M` / `MiB` ( $1,024^2$ ), `G` / `GiB` ( $1,024^3$ ), `T` / `TiB` ( $1,024^4$ ).
- **Decimal Suffixes (powers of 1000):** `kB` ( $1,000$ ), `MB` ( $1,000^2$ ), `GB` ( $1,000^3$ ), `TB` ( $1,000^4$ ).
- **Byte Suffixes:** Any suffix ending in `B` (e.g. `B`, `Bytes`, `KiB`, `MiB`, `GB`) automatically multiplies by 8 when parsed as a bit size.
- **Multiplication Expressions:** Combine factors using `*` or `x` (e.g. `188*8`, `1024*1024*8`, `1920x1080*3`).

## Real-World Recipes & Examples

```
# 1. Simple unit resizing (8-bit input to 3-bit output):
bdd "8 → 3" < foo.8bit > foo.3bit

# 2. Large units, skips, slice containers, and periodic gaps:
# Skip 1 GB, slice 6-bit unit from 8-bit container starting at bit offset 2, skip 1 MiB between
# containers:
bdd "1GB : 8[2:6] + 1MiB" < large_archive.bin

# Same layout using comma slice syntax and binary GiB:
bdd "1GiB : 8[2,6] + 1MiB" < large_archive.bin

# 3. Arithmetic container sizing (188 bytes * 8 bits = 1504-bit TS packet):
bdd "1024*1024*8 : 188*8[0:32] + 100*8" < stream.ts

# 4. Full periodic container slicing and output framing (Form A):
# Skip 123b, 8b container (2b offset, 4b payload, 2b post-gap), 8b gap between containers
# → Output 5B zero prefix, pack into 8b container (2b pre-gap, 4b payload, 2b post-gap):
bdd "123 : 8[2:4] + 8 → 5B : 8[2:4]" < in.bin > out.bin

# 5. Same container layout using Form B (Physical Layout Box):
bdd "123 : [2:4:2] + 8 → 5B : [2:4:2]" < in.bin > out.bin

# 6. MPEG-TS: Extract 13-bit PID from 188-byte packet (11-bit header):
bdd "188B[11:13] → 13" < broadcast.ts > pids.bin

# 7. Output Framing: Pack raw 13-bit PIDs back into 188-byte container frames:
bdd "13 → 188B[11:13]" < pids.bin > framed.ts

# 8. Positional Tuple Patterns (Unpack two 4-bit nibbles [A, B], output in order [B, A] via --
# rearrange=1,0, repack):
bdd 4U4U 4U4U --rearrange=1,0 < in.bin > out.bin

# 9. Positional Output Pattern with Text Tuples (Pack comma-separated "1,2" pairs into 1 byte):
bdd 4U4U --input-tuples --output-hex < pairs.txt
```

All CLI flags ( `--input-raw-unit` , `--input-offset` , `--output-raw-unit` , `--output-offset` , `--output-gap` , `--output-skip-bits` , etc.) remain fully functional and can override or complement positional arguments.

### The Next Challenge: Multi-Field Structured Records

With the unified stream mapping syntax, slicing containers and framing output is fast, expressive, and concise. But every unit we extract—whether 4 bits, 16 bits, or 32 bits—is still treated as a single monolithic block of bits or an opaque integer.

In real-world data, a 16-bit or 32-bit payload is almost never just one number. A 32-bit packet payload might contain a 1-bit boolean flag, a 3-bit status opcode, a 12-bit ADC sensor reading, and a 16-bit floating-point

weight. With only units and containers, extracting those four fields would require running four separate passes over the file with different offsets.

**What is missing?** A way to unpack a single bit unit into multiple distinct, strongly-typed fields in a single pass. This is solved in Chapter 4.

## 4. The Tuple Concept & Pattern Grammar: Multi-Field Unpacking

Rather than treating a bit unit as an opaque integer, `bdd` allows dividing a unit into named fields called a **Tuple** (analogous to a structured row in a relational database):

Raw Unit: 0 0 1 1 1 0 1 1 1 (9 bits)

Pattern: 3U 1x 2u 3M

| | | |

| | |

↳ 3-bit Signed with Sign-Split → Yields TWO fields: [1 (sign), 1

(abs)]

| |

↳ 2-bit Unsigned (Bit-Reversed) → Field value: 1

|

↳ 1-bit Skip → Discarded (not in tuple)

↳ 3-bit Unsigned Integer → Field value: 1

Unpacked Tuple: (1, 1, 1, 1)

### Pattern Specifiers Reference

Every field in an `--input-pattern` or `--output-pattern` is specified as `[<bits><type>` (with optional repetition multipliers like `4*8B`, `4*B`, or `2*(4U4u)` ).

When `<bits>` is omitted, natural bit width defaults are automatically assigned:

- `B` → 8 bits (byte; identical to `8U` or `8B` in standard big-endian bit order. Conversely, `8u` reverses bit order within the byte, matching `8b` ).
- `F` / `f` → 32 bits (IEEE 754 single)
- `D` / `d` → 64 bits (IEEE 754 double)
- `H` / `h` / `Y` / `y` → 16 bits (FP16 / BF16)
- `E` / `e` / `Q` / `q` / `C` / `c` / `K` / `k` → 8 bits (FP8 / Char / Counter)
- `m` → 4 bits (NVFP4)
- `U` / `u` / `S` / `s` / `x` / `X` / `z` / `o` / `r` → 1 bit

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| CODE                                | NAME                    | DESCRIPTION                                                                                   | ALLOWED<br>BIT WIDTHS | VALID<br>CONTEXT |
|-------------------------------------|-------------------------|-----------------------------------------------------------------------------------------------|-----------------------|------------------|
| <code>nU</code> / <code>nB</code>   | Unsigned Integer / Byte | Standard big-endian unsigned integer (default 1 / 8 bits)                                     | Any positive integer  | Input / Output   |
| <code>nu</code> / <code>nb</code>   | Unsigned (Reversed)     | Unsigned integer with reversed bit order (default 1 / 8 bits)                                 | Any positive integer  | Input / Output   |
| <code>nS</code>                     | Signed Integer          | Two's complement signed integer (returns negative values; default 1 bit)                      | Any positive integer  | Input / Output   |
| <code>ns</code>                     | Signed (Reversed)       | Two's complement with reversed bit order (default 1 bit)                                      | Any positive integer  | Input / Output   |
| <code>nM</code>                     | Sign-Split Integer      | Two's complement integer split into two fields: <code>[sign, abs(val)]</code> (default 1 bit) | Any positive integer  | Input / Output   |
| <code>nm</code>                     | Sign-Split (Reversed)   | Sign-split integer with reversed bit order (default 1 bit)                                    | Any positive integer  | Input / Output   |
| <code>32F</code> / <code>32f</code> | 32-bit Float            | IEEE 754 single-precision float ( <code>f</code> reverses bits; default 32 bits)              | Exactly 32 bits       | Input / Output   |
| <code>64D</code> / <code>64d</code> | 64-bit Double           | IEEE 754 double-precision float ( <code>d</code> reverses bits; default 64 bits)              | Exactly 64 bits       | Input / Output   |
| <code>16H</code> / <code>16h</code> | Half-Precision Float    | IEEE 754 half-precision float (FP16; default 16 bits)                                         | Exactly 16 bits       | Input / Output   |
| <code>16Y</code> / <code>16y</code> | Bfloat16                | Google Brain Bfloat16 float (BF16; default 16 bits)                                           | Exactly 16 bits       | Input / Output   |
| <code>8E</code> / <code>8e</code>   | OCP FP8 (E4M3)          | Open Compute / NVIDIA Hopper FP8 E4M3FN (default 8 bits)                                      | Exactly 8 bits        | Input / Output   |
| <code>8Q</code> / <code>8q</code>   | OCP FP8 (E5M2)          | Open Compute / NVIDIA Ada FP8 E5M2 (default 8 bits)                                           | Exactly 8 bits        | Input / Output   |
| <code>6E</code> / <code>6e</code>   | OCP FP6 (E3M2)          | Open Compute Microscaling FP6 E3M2                                                            | Exactly 6 bits        | Input / Output   |
| <code>4E</code> / <code>4e</code>   | OCP / NVFP4 (E2M1)      | NVIDIA Blackwell / OCP Microscaling FP4 E2M1 (default 4 bits)                                 | Exactly 4 bits        | Input / Output   |
|                                     |                         |                                                                                               |                       |                  |

| CODE                              | NAME               | DESCRIPTION                                                                                                                                                   | ALLOWED BIT WIDTHS   | VALID CONTEXT  |
|-----------------------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------|
| <code>nC</code> / <code>nc</code> | Characters / Bytes | Raw byte/character sequence (retains byte fidelity; default 8 bits)                                                                                           | Multiples of 8 bits  | Input / Output |
| <code>nK</code> / <code>nk</code> | Sequence Counter   | Auto-incrementing unit counter. In output patterns, emits counter without consuming tuple fields. In input patterns, reads unsigned integer. (Default 8 bits) | Any positive integer | Input / Output |
| <code>nx</code> / <code>nX</code> | Skip / Discard     | Input: skips $n$ bits. Output: consumes and discards 1 tuple field without emitting bits. (Default 1 bit)                                                     | Any positive integer | Input / Output |
| <code>nz</code>                   | Fill Zeros         | Inserts $n$ constant zero bits (default 1 bit)                                                                                                                | Any positive integer | Output only    |
| <code>no</code>                   | Fill Ones          | Inserts $n$ constant one bits (default 1 bit)                                                                                                                 | Any positive integer | Output only    |
| <code>nr</code>                   | Fill Random        | Inserts $n$ random bits from <code>/dev/urandom</code> (default 1 bit)                                                                                        | Any positive integer | Output only    |

## Pattern Repetition Multipliers

Patterns support multiplier syntax for repetitive fields, arrays, and tensor weights: `COUNT*SPEC` or `COUNT*(SPEC1 SPEC2 ...)`

### IMPORTANT

A multiplier creates  $N$  **separate fields** in the tuple, **not** a single combined integer.

- `8*3U` expands to eight separate 3-bit fields: `3U3U3U3U3U3U3U3U` (producing an 8-field tuple `(f0, f1, f2, f3, f4, f5, f6, f7)`, total width 24 bits). To extract a single 24-bit integer, use `24U` instead.
- `4*8B` (or `4*B`) expands to four 8-bit byte fields: `8B8B8B8B`.
- `10*16H` expands to ten 16-bit half-precision float fields.
- `2*(4U4u)` expands to four alternating fields: `4U4u4U4u`.
- Named repetition `w:8*3U` expands to eight indexed fields: `w_0:3U, w_1:3U, ..., w_7:3U`.

## Distinction Between `s` and `M`

- `s` (Signed Integer)**: Directly represents negative values (e.g., `-5`). When printed via `--output-tuples`, it outputs `-5`.

- **M (Magnitude & Sign Split)**: Decodes a signed two's complement number into **two separate unsigned fields**:
  1. `sign` : 0 for positive, 1 for negative.
  2. `magnitude` : `abs(value)` . This allows arithmetic, filtering, or routing based on sign and absolute value independently.

## Floating-Point Format Transcoding & The Universal `f64` Currency

`bdd` provides native bitstream transcoding between all IEEE 754 and modern AI/GPU floating-point formats:

- **64-bit Double** ( `64D` / `64d` ): IEEE 754 binary64
- **32-bit Float** ( `32F` / `32f` ): IEEE 754 binary32
- **16-bit Half** ( `16H` / `16h` ): IEEE 754 binary16 (FP16)
- **16-bit Bfloat16** ( `16Y` / `16y` ): Google Brain Bfloat16 (BF16)
- **8-bit FP8 E4M3** ( `8E` / `8e` ): OCP / NVIDIA Hopper FP8 E4M3FN (1-4-3, bias 7)
- **8-bit FP8 E5M2** ( `8Q` / `8q` ): OCP / NVIDIA Ada FP8 E5M2 (1-5-2, bias 15)
- **6-bit FP6 E3M2** ( `6E` / `6e` ): OCP Microscaling FP6 E3M2 (1-3-2, bias 3)
- **4-bit FP4 E2M1** ( `4E` / `4e` ): NVIDIA Blackwell / OCP Microscaling NVFP4 (1-2-1, bias 1)

## The `f64` Pipeline Architecture

1. **Universal Decoding to `f64`** : When reading bitstreams through `--input-pattern` , any floating-point token ( `32F` , `64D` , `16H` , `16Y` , `8E` , `8Q` , `6E` , `4E` ) is decoded into an IEEE 754 64-bit double ( `f64` ) in the internal tuple representation ( `Field::Float(f64)` ). Subnormal numbers (gradual underflow), infinities, and NaNs are decoded according to IEEE 754 and OCP specifications.
2. **Intermediate Manipulation & Rounding**: Inside the tuple pipeline, floating-point values can be rearranged, filtered, or rounded using `--round` (e.g. `--round 0,round_ties_even` , `--round 0,floor` , `--round 0,ceil` , `--round 0,trunc` ).
3. **Universal Encoding / Downcasting from `f64`** : When packing tuples into an output bitstream through `--output-pattern` , each target float token encodes the internal `f64` into its target bit layout using round-to-nearest-even (or round-to-nearest).

### TIP

**Bit-Exact Copying Without `f64` Conversion** If you want to verify or guarantee that floating-point numbers copy bit-for-bit verbatim *as is* without any possible rounding errors to/from `f64` (preserving exact signaling/quiet NaN payloads and denormals), **handle them as unsigned integers** of the corresponding bit width:

- `32U` instead of `32F` (for 32-bit single-precision floats)
- `64U` instead of `64D` (for 64-bit double-precision floats)
- `16U` instead of `16H` or `16Y` (for 16-bit FP16 or BF16)
- `8U` instead of `8E` or `8Q` (for 8-bit FP8) Unsigned integer fields ( `U` ) transfer raw bits directly with zero arithmetic translation or floating-point rounding.

## Zero-Dependency Pure-Rust AI Codecs

Standard Rust ( `std` ) only supports `f32` and `f64` natively. Experimental features ( `f16` / `f128` ) are unstable, and standard Rust provides no primitives for OCP FP8, FP6, or Blackwell FP4. `bdd` avoids heavy or unstable external dependencies by implementing custom, bit-exact codecs directly in `src/float_types.rs` , ensuring maximum performance and portability.

## Transcoding Examples

Convert between arbitrary float representations simply by combining `--input-pattern` and `--output-pattern` :

```
# Transcode 32-bit float to 16-bit IEEE half-precision (FP32 → FP16):
bdd --input-pattern=32F --output-pattern=16H < weights_fp32.bin > weights_fp16.bin

# Quantize FP16 weights directly to NVIDIA Hopper OCP FP8 (E4M3):
bdd --input-pattern=16H --output-pattern=8E < model_fp16.bin > model_fp8.bin

# Quantize FP16 to NVIDIA Blackwell 4-bit float (NVFP4 E2M1):
bdd --input-pattern=16H --output-pattern=4E < model_fp16.bin > model_fp4.bin

# Upcast Blackwell 4-bit floats back to FP32 single precision:
bdd --input-pattern=4E --output-pattern=32F < model_fp4.bin > unpacked_fp32.bin

# Quantize with explicit downward truncation before packing:
bdd --input-pattern=32F --round=0,floor --output-pattern=8E < in.bin > out.bin
```

## The Next Challenge: In-Flight Data Transformation

Pattern strings allow extracting structured multi-field tuples ( `flag`, `opcode`, `reading`, `weight` ) in a single pass. But real-world binary data is rarely in the exact format needed for downstream consumers:

- A raw sensor reading may require calibration arithmetic (e.g. multiply by 2 and add 10).
- An out-of-range sensor value may need bounding or clamping (saturation or wrapping).
- Protocol opcodes may require bitwise masking ( `AND` , `XOR` , `OR` , `NOT` ).
- Fields may need reordering, dropping, or selective filtering based on a predicate.

Under conventional Unix workflows, you would have to pipe raw binary into custom Python or C scripts to perform these simple transformations.

**What is missing?** A fast, built-in pipeline to manipulate, calculate, clamp, and filter tuple fields directly on the command line. This is solved in Chapter 5.

## 5. Flexible Pipeline & Tuple Manipulation: Transforming Data in Flight

Once unpacked into a tuple, fields can be transformed using pipeline manipulators executed in the exact order specified on the command line. Fields are referenced by index starting from `0` (or negative index from end):

### Pipeline Manipulators

- `--rearrange=F0,F1, ...` : Assembles the output tuple by listing the input field indices in the exact order you want them in the result. **This is not an imperative sequence of swap actions; it directly defines the output field order:**
  - Each entry in the comma-separated list specifies *which input field index* is placed into that slot of the result tuple.
  - Given an input tuple with 3 fields `[A, B, C]` (indices `0, 1, 2`):
    - `--rearrange=1,0,2` → `[B, A, C]` (*slot 0 gets input field 1, slot 1 gets field 0, slot 2 gets field 2*)
    - `--rearrange=2,1,0` → `[C, B, A]` (*reverses the fields*)
    - `--rearrange=1,0` → `[B, A]` (*drops field 2 entirely*)
    - `--rearrange=0,0,1` → `[A, A, B]` (*duplicates field 0*)
    - `--rearrange=-1` → `[C]` (*negative indices count backwards from the end: -1 is last, -2 second-to-last*)
- `--round=FIELD,LIMIT[,MODE]` or `--round=FIELD,MODE` (*alias: `--cut-maxint`*): Handles either upper-end range overflow (clamping/saturation/wrapping when `LIMIT` is specified) or lower-end precision reduction (floating-point rounding when `MODE` is specified). `MODE` can be specified using comma or colon (e.g. `--round 0,floor` or `--round 0,127,wrap`). When `LIMIT` is omitted, the rounding mode is applied without magnitude clamping. Supported modes:
  - `saturate` / `clamp` (default when limit is specified): Clamps out-of-bounds values to `LIMIT` or `-LIMIT`.
  - `wrap` / `wrapping`: Wraps values around using modular arithmetic (`[0, LIMIT]` for unsigned, `[-LIMIT, LIMIT]` for signed).
  - `zero` / `reset`: Sets out-of-bounds values to 0.
  - `drop` / `filter` / `checked`: Discards the tuple entirely if the value exceeds bounds.
  - `trunc` / `truncate`: Truncates magnitude toward zero.
  - `floor`: Rounds toward negative infinity.
  - `ceil`: Rounds toward positive infinity.
  - `round`: Rounds to nearest neighbor, ties away from zero.
  - `round_ties_even` / `even` / `bankers`: Rounds to nearest neighbor, ties to nearest even digit (Rust `f64::round_ties_even` / IEEE 754 default).
- `--shift-left=FIELD,BITS`: Bitwise left-shifts field by `BITS`.
- `--xor=FIELD,PARAM`: Bitwise XOR with parameter (supports hex `0x...`, bin `0b...`, or bit-count mask).

- `--and=FIELD,PARAM` : Bitwise AND with parameter.
- `--or=FIELD,PARAM` : Bitwise OR with parameter.
- `--not=FIELD` : Bitwise NOT (inverts field bits).
- `--add=FIELD,PARAM` : Adds integer parameter to field.
- `--sub=FIELD,PARAM` : Subtracts integer parameter from field.
- `--mul=FIELD,PARAM` : Multiplies field by parameter.
- `--div=FIELD,PARAM` : Integer division of field by parameter.
- `--mod=FIELD,PARAM` : Modulo of field by parameter.
- `--abs=FIELD` : Converts a signed field to its absolute value (syntax: `--abs 0` or legacy `--abs 0,0` ).
- `--sign=FIELD` : Isolates sign bit ( `0` = positive, `1` = negative).
- `--filter=FIELD,OP,VALUE` : Drops tuples where predicate is false ( `=` , `≠` , `<` , `≤` , `>` , `≥` ).

## High Bits vs. Low Bits: Range Overflow vs. Precision Rounding

When fitting a value into a smaller unit or target representation, data can be cut from two opposite ends:

| DIMENSION               | PROBLEM                                                                                  | BITS AFFECTED                 | OPERATION NAME                                        | TYPICAL COMMAND                                                            |
|-------------------------|------------------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------|----------------------------------------------------------------------------|
| Upper Bound (Range)     | Value too large to fit in container (e.g. integer 300 in an 8-bit unsigned unit max 255) | Most Significant Bits (MSBs)  | Range Clamping / Saturation / Wrapping / High-bit Cut | <code>--cut-maxint=0,255,saturate</code> or <code>-round=0,255,wrap</code> |
| Lower Bound (Precision) | Reducing fractional digits or discarding sub-unit bit resolution                         | Least Significant Bits (LSBs) | True Rounding / Quantization / Low-bit Truncation     | <code>--round=0,round_ties_even</code> or <code>--remove-right=0,4</code>  |

### 1. Upper End: Range Overflow Handling ( `--cut-maxint=FIELD,LIMIT,MODE` )

When a number is too large to fit in a target container, its most significant bits must be cut or handled. **This is not rounding; it is range bounding, clipping, or saturation:**

- `saturate` / `clamp` (default): Clips values exceeding `LIMIT` down to `LIMIT` (e.g., 300 becomes 255).

- **wrap / wrapping** : Truncates high-order bits, wrapping modulo the container capacity (e.g.,  $300 \pmod{256} = 44$ ).
- **drop / checked** : Rejects and discards the tuple completely if out of bounds.
- **zero / reset** : Resets out-of-range values to zero.

## 2. Lower End: True Rounding & Precision Reduction ( `--round=FIELD,MODE` )

When reducing precision or discarding fractional parts, bits are removed from the **least significant part**:

- **Floating-point rounding**: `--round=0,floor` , `--round=0,ceil` , `--round=0,trunc` , `--round=0,round` , or `--round=0,round_ties_even` rounds fractional values to integers according to IEEE 754 rules.
- **Integer bit shifting / truncation**: `--remove-right=0,N` bitwise shifts right by  $N$  bits, discarding the  $N$  least significant bits (integer truncation towards zero).
- **Integer true rounding (half-up)**: To round an integer to the nearest multiple of  $2^N$  instead of truncating it, add a rounding bias of half the divisor before shifting right:

```
# Divide field 0 by 16 (2^4) with round-to-nearest (half-up) instead of truncation:
bdd --add 0,8 --remove-right 0,4
```

## Ordered Pipeline Example

Chain multiple transformations in arbitrary sequence:

```
# Add 10, multiply by 2, and filter values greater than 25:
bdd --input-counter --count 10 --input-pattern 8U \
    --add 0,10 --mul 0,2 --filter "0,>,25" --output-json
```

## Advanced Bit-Slicing Recipe

*Problem from presentation:* Input is a stream of signed bytes ( `8M` ). We want to clamp the value so  $-128$  becomes  $-127$ , isolate the magnitude, and extract only the most significant bit of the magnitude:

```
bdd --input-pattern=8M --rearrange=-1 --cut-maxint=0,127 --remove-right=0,6 --output-unit=1 <
input.bin
```

Explanation:

1. `8M` produces `[sign, magnitude]` .
2. `--rearrange=-1` selects only the last field (magnitude).
3. `--cut-maxint=0,127` clamps the magnitude to 127.
4. `--remove-right=0,6` shifts right by 6 bits, leaving the 7th bit (MSB of magnitude).
5. `--output-unit=1` outputs the single bit.

## The Next Challenge: Visualization, Pipeline Sinks & Test Data

We can now slice, unpack, and manipulate bitstreams in flight. But where does the output go? If `bdd` only writes raw binary bytes, inspecting unaligned fields or verifying single-bit transformations on a terminal is difficult.

Furthermore, how do you pipe structured tuples into standard Unix tools like `jq`, `awk`, Python, or relational databases? And what if you don't have an input file at all, but need to generate synthetic bitstreams (sequential counters, cryptographically strong random noise, endless zeros) to test an algorithm, or ingest comma-separated text files to pack custom binary files?

**What is missing?** Diverse human-readable and machine-readable output sinks, synthetic stream generators, and text tuple ingestion. This is solved in Chapter 6.

---

## 6. Sinks, Synthetic Streams & Text Formats: Interfacing with the World

---

`bdd` can synthesize bitstreams without requiring input files, and output in human-readable or script-friendly formats:

### Synthetic Stream Sources

- `--input-zeros ( -0 )`: Endless stream of zero bits.
- `--input-ones ( -1 )`: Endless stream of one bits.
- `--input-random ( -r )`: Random bits from `/dev/urandom` (cryptographically strong OS randomness, not PRNG).
- `--input-counter ( -c )`: Sequential integer counter starting at 0, incrementing by 1 per unit.
- `--count=N`: Limit processing to  $N$  units (0 = infinite).
- `--skip=N`: Skip  $N$  initial units before processing.

### Human-Readable & Script Sinks

- `--output-hex ( -x )`: Formatted hexadecimal representation per unit, grouped neatly into power-of-two line widths.
- `--output-bits ( -b / --output-bit )`: ASCII bit strings ( `'0'` and `'1'` ) per unit.
- `--output-integers ( -i / --output-integer )`: One unsigned decimal integer per line (ideal for Unix pipelines: `awk`, `sort`, `uniq` ).
- `--output-tuples ( -T / --output-tuple )`: Comma-separated values per unit.
- `--output-json`: Newline-delimited JSON (NDJSON) record per tuple (ideal for `jq`, Python, databases).
- `--output-csv`: Comma-separated values (with optional `--csv-header="col1,col2"` ).
- `--output-visual`: Interactive colorized terminal dump rendering unaligned field slices in alternating ANSI colors.
- `--input-tuples ( -t )`: Ingest comma-separated values directly from stdin/file into tuples. Quoted values (e.g. `"hello"`, `'123'` ) are preserved as byte strings ( `Field::Bytes` ) instead of numbers,

preserving exact text for character/byte patterns ( `c` / `C` ).

## Packing from Text & Tuples

```
# Pack octal numbers from text tuples into binary:
printf "1,2,3\n3,7,7\n" | bdd --input-tuples --output-pattern='2U3U3U' | od -t o1
# Outputs: 123 377

# Pack two 4-bit numbers into hex bytes:
echo -en "1,2\n3,4" | bdd --input-tuples --output-pattern=4U4U | od -t x1
# Outputs: 12 34

# Discard tuple fields on output using 'x' (drops the second field):
echo -en "10,99,20\n" | bdd --input-tuples --output-pattern='8U,x,8U' --output-hex
# Outputs: 0a14

# Inject an embedded auto-incrementing sequence counter using 'k' / 'K':
echo -en "10,20\n30,40\n" | bdd --input-tuples --output-pattern='8U,8U,8K' --output-hex
# Outputs: 0a1400 1e2801

# Pack bare pattern types with natural default widths:
printf "1,2\n" | bdd --input-tuples --output-pattern='BB' --output-hex
# Outputs: 0102

# Pack text numbers into 64-bit IEEE double-precision floats:
printf "1.0\n2.0\n" | bdd --input-tuples --output-pattern=64D > doubles.bin
```

## The Next Challenge: Multi-Stream Coordination

So far, all operations have acted on a single input stream, producing a single output stream. But real-world data pipelines rarely operate in isolation.

Consider audio processing (interleaving Left and Right channels into stereo), system telemetry (multiplexing a hardware timestamp counter into an existing sensor feed), or packet processing (splitting multiplexed transport streams into separate audio, video, and subtitle files). If you have two independent binary streams, how do you interleave them into one? And conversely, how do you split a multi-field tuple into dedicated destination files without running multiple passes?

**What is missing?** Symmetrical multi-stream coordination: interleaving secondary streams round-robin into primary data, and demuxing tuple fields into separate files. This is solved in Chapter 7.

---

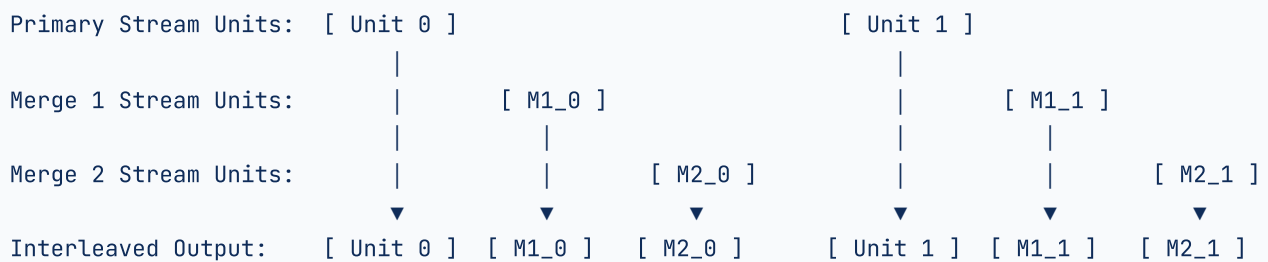
## 7. Stream Merging, Interleaving & Demuxing: Multi-Channel Coordination

---

When processing multi-channel systems, `bdd` provides symmetrical stream merging (combining multiple streams into one) and field demuxing (routing fields to separate destination files or pipes).

## Stream Merging & Round-Robin Interleaving

The merge stream reads a secondary file and interleaves its data into the primary stream:



- `--merge-file=PATH` : Interleave data from secondary files (can be specified multiple times for N-way round-robin merge, "-" for stdin).
- `--merge-files=PATHS` : Comma- or space-separated list of merge files to interleave round-robin.
- `--merge-unit=BITS` : Size of merge units in bits (default: 8).
- `--merge-copy-first=BITS` : Copies an initial bit header/preamble from the merge file *before* starting the interleaved loop.
- `--drop-partial-eof` (aliases: `--drop-trailing-bits`, `--no-pad-eof`) : Discards incomplete trailing bits at EOF instead of zero-padding them into a synthetic extra unit.
- `--merge-drop-partial-eof` : Discards incomplete trailing bits at EOF in the merge stream.

### Example: Hex Dump with Interleaved Memory Addresses

Interleave a 12-bit linear address counter with actual data bytes from `/etc/passwd` :

```
bdd --input-counter --count=16 --input-unit=12 --output-unit=12 --merge-file=/etc/passwd --output-hex
```

Output:

```
000 72 001 6f 002 6f 003 74 004 3a 005 78 006 3a 007 30
008 3a 009 30 00a 3a 00b 72 00c 6f 00d 6f 00e 74 00f 3a
```

### Example: Multi-File Round-Robin Interleaving

Interleave primary stream units with units from two independent secondary streams:

```
bdd --input-counter --count=2 --input-unit=8 --output-unit=8 \
    --merge-file=ch1.bin --merge-file=ch2.bin --output-hex
# or equivalently:
bdd --input-counter --count=2 --input-unit=8 --output-unit=8 \
    --merge-files=ch1.bin,ch2.bin --output-hex
```

## Channel Demuxing & Splitting

When processing multiplexed packet formats or interleaved bitstreams (e.g. audio + video, header + payload), `bdd` can demux fields into independent files or pipes:

```
# Split interleaved 16-bit audio and 32-bit video into separate streams:
bdd --input-pattern=16B32B \
    --demux 0:audio.raw \
    --demux 1:video.raw < input.bin > /dev/null

# Alternatively, using --demux-files shorthand:
bdd --input-pattern=16B32B --demux-files=audio.raw,video.raw < input.bin > /dev/null
```

## The Next Challenge: Endianness and Bit-Order Reversals

Now we can slice containers, unpack tuples, transform values, and coordinate multiple streams. But what happens when bitstreams originate from different hardware architectures or transmission protocols?

A microcontroller transmitting over SPI or UART often sends the least significant bit (LSB) first, while network protocols and standard file formats expect most significant bit (MSB) first. Little-endian and big-endian integers swap bytes, but what if bits within each byte, or bits across an arbitrary 12-bit unit, need to be reversed? Attempting to reverse bits using ad-hoc shifts or bitwise operations in shell scripts leads to pervasive confusion and subtle bugs.

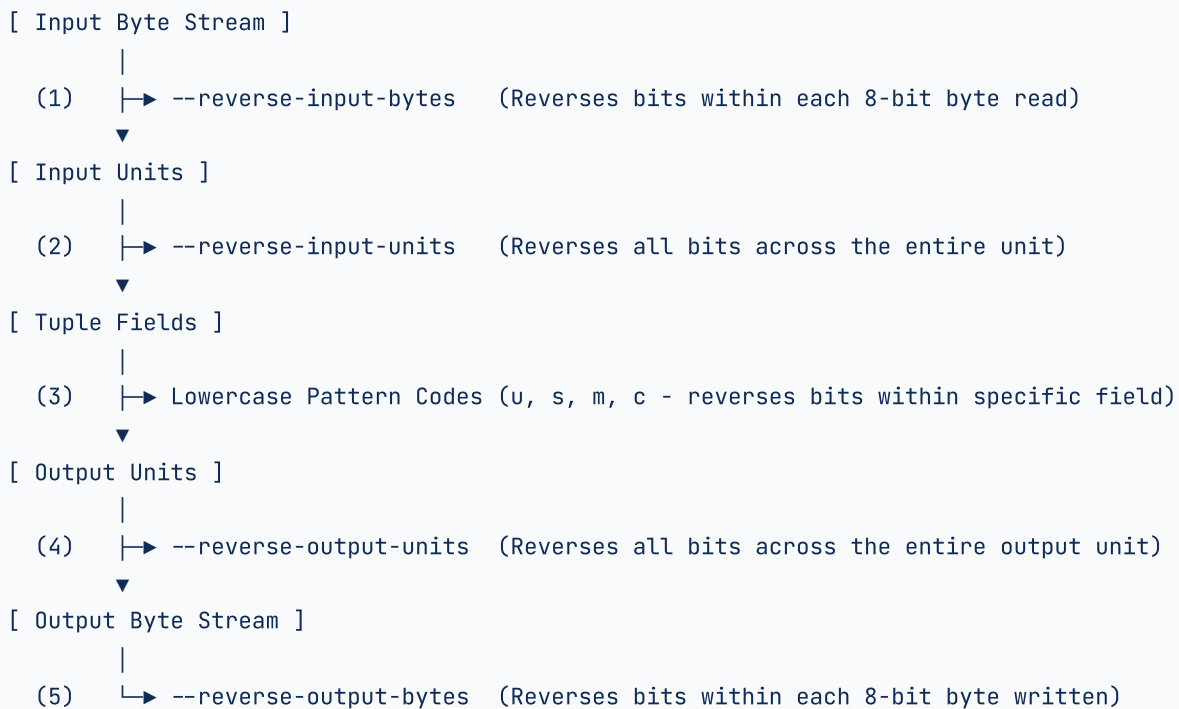
**What is missing?** A clean, layered architecture that isolates bit reversals into predictable stages. This is solved in Chapter 8.

---

## 8. De-mystifying Bit Reversals: Taming Endianness and Bit Order

---

Reversing bits in binary processing can easily become confusing. `bdd` cleanly isolates bit reversals into four distinct operational layers:



- `--input-little-endian` : Shorthand combining `--reverse-input-bytes` and `--reverse-input-units` .
- `--output-little-endian` : Shorthand combining `--reverse-output-bytes` and `--reverse-output-units` .

## The Next Challenge: Gigabyte Skips, Hardware Seeking & Large Dimensions

Our operations are precise down to the single bit. But how does this scale when working with multi-gigabyte disk images, massive uncompressed 4K/8K video frames, or high-throughput satellite telemetry?

Does skipping 10 GB into a file require reading 10 billion bytes through RAM? And how do you comfortably express large dimensions like `1920x1080x24` or 10 GiB on the command line without writing out twelve trailing zeros?

**What is missing?** Fast  $O(1)$  hardware filesystem seeking, an exabyte-scale 64-bit architecture, size suffixes ( `GiB` , `MB` , `B` ), and multiplication expressions ( `1920x1080x24` ). This is solved in Chapter 9.

## 9. Exabyte Scale, Fast Hardware Seeking & Size Expressions

Can `bdd` skip over gigabytes of data on the command line? **Yes, without limit:**

- **Full 64-Bit Architecture ( `u64` )**: All skips, offsets, gaps, units, and record counts are represented internally as 64-bit unsigned integers. `bdd` can represent bit offsets up to  $2^{64} - 1 \approx 1.84 \times 10^{19}$  bits, which equals **2.3 Exabytes** (2,305,843,009 Gigabytes).
- **Filesystem Seek Limits**: On seekable files, Linux `lseek64` supports offsets up to  $2^{63} - 1$  bytes = **9.22 Exabytes**, executed in  $O(1)$  constant time (microseconds) without memory overhead.

- **Human-Friendly Size Suffixes:** You do not need to calculate zeroes or bit multiplications manually. All numeric arguments accept standard scale suffixes:
  - **Binary multiples (powers of 1024):** K, M, G, T, P, E (or Ki, Mi, Gi, Ti, Pi, Ei).
  - **Decimal multiples (powers of 1000):** KB, MB, GB, TB, PB, EB.
  - **Byte-scaled bit skips:** On bit options ( `--input-skip-bits` , `--merge-skip-bits` ), specifying B (e.g. 10GiB, 4GB, 100B) automatically multiplies bytes by 8 bits.
- **Multiplication Expressions:** Numbers on the command line never run out or require manual mental calculations. All size and count options support multiplication expressions using `*` or `x` :
  - **Unit-stride skipping:** Skip 1,000,000 24-bit units directly: `--input-skip-bits=1000000*24` (or `--input-skip-bits=1000*1000*24` or `--input-skip-bits=1M*24` ).
  - **2D/3D dimensions and frame strides:** Easily express multi-dimensional buffers: `--input-skip-bits=1920x1080*24` or `--input-skip-bits="1920 * 1080 * 3B"` .
  - **Sizing units and offsets:** Express byte multiples: `--input-unit=3*8` (24-bit unit) or `--input-offset=2*8` (16-bit offset).
  - **Scaled counts and repeats:** `--count=1000*10` , `--input-repeat=10*5` .

```
# Skip 1,000,000 24-bit units directly in bits:
bdd --input-file=samples.bin --input-skip-bits=1000000*24 --output-hex

# Skip a 1080p 24-bit uncompressed RGB frame buffer:
bdd --input-file=video.raw --input-skip-bits=1920x1080*24 --count=100 --output-hex

# Instantly seek 10 GiB into a file and extract 4 bytes in hex:
bdd --input-file=large_disk.img --input-skip-bits=10GiB --count=4 --output-hex

# Skip 10 million 8-bit units:
bdd --input-file=stream.bin --input-unit=8 --input-skip-units=10M < stream.bin

# Process at most 500k records:
bdd --input-tuples --count=500k < records.csv
```

## Fast Hardware Seeking vs Streaming Fallback

Whenever the input stream is a seekable regular file or file descriptor, `bdd` automatically invokes  $O(1)$  filesystem `lseek64` to skip over initial skips, gaps, and raw unit post-gaps. This allows skipping past 50 GB of data in microseconds without consuming disk I/O bandwidth or RAM.

If the input is non-seekable (such as a standard pipe `cat file | bdd`, FIFO, or network socket), `bdd` automatically falls back to sequential streaming reads, discarding skipped bits seamlessly. Seeking can be explicitly disabled on seekable files using `--no-seek` (or `--input-no-seek`, `--merge-no-seek`) to benchmark pure stream throughput.

## The Next Challenge: Discovery & Inspection of Unknown Formats

Everything so far assumes you already know the exact schema, stride, and byte layout of your binary stream.

But what if you encounter an unfamiliar proprietary file format, a corrupted firmware image, or an unknown radio capture? How do you discover its periodic stride, byte distribution, or Shannon entropy without guessing? And in modern workflows, how can autonomous AI coding agents, reverse engineers, or visual analysts interact with `bdd` through standard protocols or interactive web tools?

**What is missing?** Automated binary probing, Shannon entropy analysis, schema explanation, standard presets, Model Context Protocol (MCP) server support, and an interactive Web GUI. This is solved in Chapter 10.

---

## 10. AI Ergonomics, Binary Inspection & Web GUI

---

`bdd` provides native primitives for autonomous AI coding agents, reverse engineers, and pipeline automation:

### Built-in Format Presets ( `--preset` , `--list-presets` )

Instead of manually calculating complex bit offsets, standard protocol and AI weight presets configure input patterns, unit widths, and field names in one step:

```
# List all 13 standard presets:
bdd --list-presets
```

Standard presets include:

- `mp3-header` (MPEG Audio Frame Header, 32 bits)
- `mpeg-ts` (MPEG Transport Stream Header, 32 bits from 188B packet)
- `wav-header` (RIFF WAV Header Identifier, 96 bits)
- `jpeg-sof0` (JPEG Start of Frame 0, 80 bits)
- `h264-nal` (H.264 / AVC NAL Unit Header, 8 bits)
- `nvfp4` (Dual packed NVIDIA NVFP4 E2M1 weights, 8 bits)
- `fp6-e3m2` (Quad packed FP6 E3M2 AI weights, 24 bits)
- `fp8-e4m3` (OCP FP8 E4M3 AI weight, 8 bits)
- `fp8-e5m2` (OCP FP8 E5M2 AI weight, 8 bits)
- `bf16` (Bfloat16 Brain Floating Point, 16 bits)
- `fp16` (IEEE 754 Half-Precision Float, 16 bits)
- `ipv4-header` (IPv4 Packet Header, 160 bits / 20 bytes)
- `udp-header` (UDP Datagram Header, 64 bits / 8 bytes)
- `tcp-header` (TCP Segment Header with discrete sub-byte flags, 160 bits / 20 bytes)
- `riscv-r-type` (RISC-V 32-bit R-type Instruction, 32 bits)

### Keyed JSON Objects ( `--json-object` , `--json-fields` )

Pair named patterns or presets with `--json-object` to emit newline-delimited JSON dictionaries where keys match field names:

```
bdd --input-file stream.ts --input-raw-unit 188B --preset mpeg-ts --json-object --count 3
```

Output:

```
{"afc":1,"cc":0,"pid":0,"priority":0,"pusi":0,"scrambling":0,"sync":71,"tei":0}
{"afc":1,"cc":1,"pid":17,"priority":0,"pusi":1,"scrambling":0,"sync":71,"tei":0}
{"afc":1,"cc":2,"pid":256,"priority":0,"pusi":1,"scrambling":0,"sync":71,"tei":0}
```

## Pattern Explainer ( `--explain-pattern` )

Examine bit ranges, byte alignments, offsets, and field types without running a processing job:

```
bdd --explain-pattern "sync:11u,version:2u,layer:2u,protect:1b,bitrate:4u"
# Machine-readable JSON output for AI toolchains:
bdd --explain-pattern "sync:11u,version:2u" --output-json
```

## Binary Prober ( `--probe` )

Inspect unknown binary blobs without prior schema knowledge. Computes Shannon entropy ( $H = -\sum p_i \log_2 p_i$ ), byte class distributions, periodic stride autocorrelation across offsets 1..512 bytes (detecting MPEG-TS, 24-bit audio, or fixed-stride telemetry), and extracts printable ASCII strings:

```
bdd --probe payload.bin
# Full JSON metrics:
bdd --probe payload.bin --output-json
```

## Model Context Protocol (MCP) Server ( `--mcp` )

`bdd` includes a native JSON-RPC 2.0 stdio MCP server for agent integration:

```
bdd --mcp
```

Registered MCP tools:

- `bdd_slice` : Slices a file or hex string by pattern/preset and outputs text or JSON.
- `bdd_probe` : Analyzes entropy, periodic strides, and format heuristics.
- `bdd_explain_pattern` : Explains schema bit offsets and field types.
- `bdd_list_presets` : Returns available protocol presets.

## Interactive Web Application ( `--serve` )

`bdd` embeds a complete, zero-dependency browser application directly into the binary:

```
bdd --serve          # Launches web UI at http://localhost:7788
bdd --serve 8080      # Custom port (aliases: --web, --gui)
# Or via make targets:
make web             # Compiles and runs ./bdd --serve
make web-py          # Standalone Python backend (python3 web/server.py)
```

### Key Features:

- **File Upload & Drag-and-Drop:** Upload binary blobs, capture files, or sample datasets.
- **Protocol Presets:** 1-click loading for MPEG-TS, MP3 Header, NVFP4, FP8, WAV, RGB565, and more.
- **Live Visual Bit Breakdown:** Color-coded field layout (uint, int, float, char, discard, counter) with exact bit offsets and byte boundaries.
- **Real-Time Command Generator:** Generates equivalent copy-pasteable `bdd` CLI commands as you tune options.
- **Multi-Sink Results Inspector:** Switch seamlessly between Formatted JSON, Hex Dump, Color Matrix, CSV, Raw Bits, and Binary Download.
- **Binary Prober in Browser:** Run entropy, byte class, and periodic stride autocorrelation with one click.

### Agent Documentation ( `llms.txt` & Agent Skill)

- `llms.txt` : High-density reference tailored for LLM context windows.
- **Agent Skill:** Available at `&approx;/.agents/skills/bdd/SKILL.md` .

### The Next Challenge: Direct Programmatic Integration

The CLI, interactive Web GUI, and MCP server cover exploration, scripts, and autonomous AI agents. But what if you need to embed `bdd` 's ultra-fast bitstream engine directly into your own high-performance C, C++, or Python applications without spawning shell subprocesses?

**What is missing?** Clean, native programmatic C-ABI headers and zero-dependency Python wrappers. This is solved in Chapter 11.

---

## 11. Programmatic Interfaces: C API and Python SDK

---

`bdd` exports clean C-ABI symbols in `libbdd.so` and includes an official C header ( `include/bdd.h` ) and a zero-dependency Python wrapper ( `python/bdd.py` ):

## Python ( ctypes & pyproject.toml )

```
from bdd import Bdd

b = Bdd()

# Fast hardware bit-reversal:
assert b.reverse_bits(0x0F, 8) == 0xF0

# Unpack bitfields into Python tuple:
fields = b.unpack("4U4U", 0xA5) # → (10, 5)

# Pack Python tuple into raw integer:
unit = b.pack("4U4U", (10, 5)) # → 0xA5

# Native AI float conversion (FP16, BF16, FP8, FP4):
val = b.decode_f16(0x3C00) # → 1.0
bits = b.encode_f16(1.0) # → 0x3C00
```

## C / C++ API

```
#include "bdd.h"
#include <stdio.h>

int main() {
    uint64_t fields[2];
    bdd_unpack_u64("4U4U", 0xA5, fields, 2);
    printf("Field 0: %lu, Field 1: %lu\n", fields[0], fields[1]); // 10, 5
    return 0;
}
```

## The Next Challenge: Engine Architecture & Performance Guarantees

How does `bdd` achieve multi-gigabit throughput across arbitrary non-byte-aligned boundaries? What are the architectural guarantees regarding UTF-8 string integrity, zero-overhead diagnostic logging, and bignum arithmetic?

**What is missing?** Detailed performance benchmarks, architectural invariants, and UTF-8 stream processing analysis. This is documented in Chapter 12.

---

## 12. Architecture, Performance & Benchmarks

`bdd` achieves high throughput across arbitrary bit boundaries, balancing hardware register acceleration for sub-64-bit units with arbitrary-precision arithmetic for large bignum fields.

## Throughput Benchmarks

Measured via `make bench` ( `benches/throughput.rs` ) on Linux x86\_64:

| OPERATION                                         | TOTAL VOLUME | THROUGHPUT (BITS/S) | THROUGHPUT (BYTES/S) | NOTES                                          |
|---------------------------------------------------|--------------|---------------------|----------------------|------------------------------------------------|
| Hardware 64-bit Bit Reversal                      | 128 Mbits    | 16.00 Gbps          | 2,000 MB/s           | Direct CPU<br><code>u64::reverse_bits()</code> |
| Bignum 1024-bit Packing/Unpacking                 | 10.2 Mbits   | 5.27 Gbps           | 658.6 MB/s           | Large-block bignum bitfield packing            |
| Bignum 256-bit Packing/Unpacking                  | 12.8 Mbits   | 1.90 Gbps           | 237.2 MB/s           | SHA-256 size field packing/unpacking           |
| Synthetic 8-bit Linear Stream                     | 16.0 Mbits   | 234.5 Mbps          | 29.3 MB/s            | Continuous bit generation & sink flush         |
| 1-bit Single-Bit Resolution Stream                | 0.5 Mbits    | 43.3 Mbps           | 5.4 MB/s             | Single-bit slice accumulation & packing        |
| Bignum 1024-bit Bit Reversal                      | 5.1 Mbits    | 39.4 Mbps           | 4.9 MB/s             | Full arbitrary-precision bit reversal          |
| Unaligned 3-bit to 8-bit Extraction               | 3.0 Mbits    | 39.0 Mbps           | 4.9 MB/s             | Cross-byte boundary accumulation               |
| Tuple Pipeline ( <code>2U3U3U</code> → Rearrange) | 2.0 Mbits    | 29.7 Mbps           | 3.7 MB/s             | Multi-field unpack, reorder & repack           |

## UTF-8 Stream Processing & Architectural Study

For an in-depth analysis of UTF-8 bitstream hazards, byte-alignment constraints, continuation header preservation, and Unicode scalar value processing in `bdd`, consult the technical report:

- [docs/utf8\\_study.md](#)

## Codebase Organization

The Rust implementation is organized cleanly into modular crates:

```

src/
├─ lib.rs          # Public library crate interface
├─ main.rs         # Thin CLI wrapper & early dispatcher
├─ error.rs        # Strongly-typed BddError hierarchy
├─ diag.rs         # High-throughput warning deduplication & diagnostic reporting
├─ counter.rs      # Unit and skip counting logic
├─ field.rs        # Arbitrary-precision Field enum & hardware bit-reversals
├─ float_types.rs  # AI/GPU float codecs (FP16, BF16, FP8 E4M3/E5M2, FP6, FP4)
├─ ffi.rs          # C-ABI export symbols for native integration
├─ pattern.rs      # Grammar parser, multipliers, TupleUnpacker & TuplePacker
├─ stream.rs       # Stream generators (File, Counter, Zeros, Ones, Random, Tuples)
├─ sink.rs         # Output writers (Binary, Hex, Bit, Integer, CSV, NDJSON, ANSI Visual)
├─ manipulator.rs  # Ordered pipeline transformations (Arithmetic, Bitwise, Filter)
├─ preset.rs       # Standard protocol & AI float presets (mp3, ts, wav, nvfp4, etc.)
├─ explain.rs      # Pattern bit layout, alignment & schema analysis
├─ probe.rs        # Shannon entropy, periodic stride autocorrelation & byte classes
├─ mcp.rs          # Native JSON-RPC 2.0 Model Context Protocol (MCP) server
├─ server.rs       # Embedded zero-dependency HTTP server & web app dispatcher
├─ cli.rs          # Clap CLI definition & validation rules
└─ engine.rs       # End-to-end pipeline execution orchestrator
include/
└─ bdd.h           # C/C++ API header
python/
├─ __init__.py     # Python package root
└─ bdd.py          # Zero-dependency Python ctypes wrapper
web/
├─ index.html      # Responsive browser single-page application UI
├─ style.css        # Dark-slate styling & color-coded bit pattern layouts
├─ app.js          # Interactive JavaScript client & preset engine
└─ server.py       # Standalone Python HTTP server
package/
├─ build_deb.sh    # Builds Debian / Ubuntu (.deb) packages with dpkg-deb
├─ build_rpm.sh    # Builds Fedora / RHEL (.rpm) packages for DNF with rpmbuild
├─ build_all.sh    # Builds all distribution packages and generates SHA256SUMS
└─ README.md       # Packaging documentation and install instructions
pyproject.toml     # Standard Python package configuration
llms.txt           # High-density agent & LLM reference card

```

## Key Design Principles

1. **Strict Type Safety:** All errors flow through `BddError`. Functions return `Result<T, BddError>` instead of panicking or calling `std::process::exit`.
2. **Fast-Path Bit Reversal:** Sub-64-bit integer bit reversals execute via direct hardware `u64::reverse_bits()`, falling back to `BigUint` bit arithmetic only when necessary.
3. **Byte-Level String Integrity:** The `C` and `c` pattern types store raw bytes internally (`Field::Bytes`) rather than lossy UTF-8 conversions, guaranteeing bit-perfect roundtrips.
4. **Automated Verification:** Integrated test runner runs native Rust unit tests, bignum tests, and legacy golden-file integration tests.

5. **Zero I/O Diagnostic Bottlenecks:** Diagnostic warnings during stream processing are deduplicated with  $O(1)$  hashing, printing on first encounter and summarizing at EOF, or completely silenced via `-q` / `--quiet` to prevent `stderr` I/O serialization from bottlenecking multi-gigabit throughput.

## The Complete Command Reference

Having explored all bitstream concepts, physical containers, pattern tuples, pipeline transformations, multi-stream coordination, bit reversals, fast hardware seeking, AI inspection tools, and programmatic APIs, Chapter 13 provides the comprehensive reference manual for all command-line options.

---

## 13. Command Line Options Reference

Usage: bdd [OPTIONS] [STREAM\_PATTERN] [INPUT\_PATTERN] [OUTPUT\_PATTERN]

### Arguments:

|                  |                                                                              |
|------------------|------------------------------------------------------------------------------|
| [STREAM_PATTERN] | Stream I/O pattern (e.g. '8→3', '123:8[2:4]+8 → 5B:8[2:4]', '[2:4:2] → 4')   |
| [INPUT_PATTERN]  | Input pattern for unpacking tuples (e.g. '8U', '24S', 'sync:11u,version:2u') |
| [OUTPUT_PATTERN] | Output pattern for packing tuples (e.g. '8U', '8U,x,8U')                     |

### Input Unit & Raw Unit Options:

|                               |                                                                         |
|-------------------------------|-------------------------------------------------------------------------|
| -p, --input-pattern <PATTERN> | Bit pattern to unpack input (e.g. "3U1x2u3M")                           |
| -u, --input-unit <BITS>       | Input unit size in bits (shorthand for <BITS>U)                         |
| --preset <NAME>               | Use built-in protocol or float preset (e.g. mp3-header, mpeg-ts, nvfp4) |
| --list-presets                | List all built-in format presets and exit                               |
| --input-skip-bits <BITS>      | Initial bit offset before first unit [default: 0]                       |
| --input-skip-units <UNITS>    | Skip initial N units from input stream [default: 0]                     |
| --input-gap <BITS>            | Bit gap skipped after each unit (or between raw units) [default: 0]     |
| --input-raw-unit <BITS>       | Size of repeating raw container/frame in bits                           |
| --input-offset <BITS>         | Bit offset of unit inside raw unit [default: 0]                         |
| --input-assert-aligned        | Error if EOF is not byte-aligned                                        |
| --drop-partial-eof            | Discard incomplete trailing bits at EOF instead of zero-padding         |
| --no-peek, --do-not-peek      | Globally disable seeking on all inputs (force streaming read)           |
| --input-no-peek               | Disable seeking specifically on primary input                           |
| --input-use-peek              | Explicitly enable seeking on input (default: true)                      |

### Synthetic Stream Sources:

|                     |                                                                |
|---------------------|----------------------------------------------------------------|
| -c, --input-counter | Generate sequential counter numbers (0, 1, 2...)               |
| -0, --input-zeros   | Generate endless stream of zero bits                           |
| -1, --input-ones    | Generate endless stream of one bits                            |
| -r, --input-random  | Generate random bits from /dev/urandom                         |
| -t, --input-tuples  | Read comma-separated tuple lines from text input               |
| --skip <UNITS>      | Skip initial N units [default: 0]                              |
| --count <COUNT>     | Process at most N units (0 or omitted = infinite) [default: 0] |

### Bit Reversal Options:

|                        |                                             |
|------------------------|---------------------------------------------|
| --reverse-input-bytes  | Reverse bit order within each input byte    |
| --reverse-input-units  | Reverse bit order across entire input unit  |
| --input-little-endian  | Combined byte and unit reversal for input   |
| --reverse-output-bytes | Reverse bit order within each output byte   |
| --reverse-output-units | Reverse bit order across entire output unit |
| --output-little-endian | Combined byte and unit reversal for output  |

### Tuple Manipulators:

```

--rearrange <FIELDS>      Explicit output field order by input index (e.g. "1,0,2",
"1,0", "-1", "0,0")
--round <F,LIMIT[,MODE]>  Round or clamp field F (alias: --cut-maxint; modes:
saturate, wrap, zero, drop, trunc, floor, ceil, round, round_ties_even)
--remove-right <F,BITS>  Right-shift field F by BITS
--shift-right <F,BITS>   Right-shift field F by BITS (synonym)
--shift-left <F,BITS>    Left-shift field F by BITS
--xor <F,PARAM>          Bitwise XOR field F with PARAM (or mask of N ones)
--and <F,PARAM>          Bitwise AND field F with PARAM
--or <F,PARAM>           Bitwise OR field F with PARAM
--not <FIELD>            Bitwise NOT (inverts bits of field F)
--add <F,PARAM>          Add PARAM to field F
--sub <F,PARAM>          Subtract PARAM from field F
--mul <F,PARAM>          Multiply field F by PARAM
--div <F,PARAM>          Integer division of field F by PARAM
--mod <F,PARAM>          Modulo of field F by PARAM
--abs <FIELD>            Replace signed field F with abs(F)
--sign <FIELD>           Replace signed field F with sign bit (0/1)
--filter <F,OP,VAL>      Filter tuples where predicate is false (=, ≠, <, ≤, >,
≥)

Output Unit & Pattern Options:
--output-pattern <PATTERN> Bit pattern to pack output (supports AI floats, multipliers
& counter)
--output-unit <BITS>       Output unit size in bits (default: 8, or pattern width)
--output-raw-unit <BITS>   Size of repeating raw unit / container in bits on output
--output-offset <BITS>    Bit offset of unit inside output raw unit [default: 0]
--output-gap <BITS>       Bit gap between output raw unit containers [default: 0]
--output-skip-bits <BITS>  Initial zero prefix bits emitted before first unit [aliases:
--output-skip, --output-prefix]
-x, --output-hex           Output units as formatted hexadecimal strings
-b, --output-bits         Output units as ASCII bit strings ('0' and '1')
-i, --output-integers     Output unsigned integer per unit (one per line)
-T, --output-tuples       Output fields as comma-separated tuples
--output-json             Output tuples as newline-delimited JSON (NDJSON)
--json-object             Emit JSON as keyed objects instead of positional arrays
--json-fields <FIELDS>   Explicit comma-separated field names for JSON/CSV
--output-csv              Output tuples as CSV
--csv-header <HEADER>    Optional CSV column header row
--output-visual           Colorized ANSI terminal dump of unaligned fields

Inspection, Web UI & Model Context Protocol (MCP):
--explain-pattern [PATTERN] Analyze bit layout, byte alignment, and field breakdown
--probe [FILE]             Inspect binary entropy, byte classes, periodic strides, and
strings
--mcp                      Launch native JSON-RPC 2.0 Model Context Protocol (MCP)
server
--serve [PORT]            Launch interactive Web UI browser app (aliases: --web, --
gui) [default: 7788]

Demuxing & Channel Splitting:
--demux <FIELD:PATH>      Route individual tuple field to a dedicated output file

```

|                                                       |                                                                    |
|-------------------------------------------------------|--------------------------------------------------------------------|
| <code>--demux-files &lt;PATHS&gt;</code>              | Comma-separated list of output files for fields 0, 1, 2...         |
| <code>--input-repeat &lt;COUNT&gt;</code>             | Repeat input stream N times (0 = infinite)                         |
| <br>Merge Options:                                    |                                                                    |
| <code>--merge-file &lt;PATH&gt;</code><br>merge)      | Interleave stream from secondary file (repeatable for N-way merge) |
| <code>--merge-files &lt;PATHS&gt;</code>              | Comma- or space-separated list of merge files                      |
| <code>--merge-unit &lt;BITS&gt;</code>                | Bit width of each merge unit [default: 8]                          |
| <code>--merge-skip-bits &lt;BITS&gt;</code>           | Initial bit offset in merge file [default: 0]                      |
| <code>--merge-skip-units &lt;UNITS&gt;</code>         | Skip initial N units in merge file [default: 0]                    |
| <code>--merge-gap &lt;BITS&gt;</code><br>[default: 0] | Bit gap skipped after each merge unit (or between raw units)       |
| <code>--merge-copy-first &lt;BITS&gt;</code>          | Copy initial header bits from merge file first                     |
| <code>--merge-raw-unit &lt;BITS&gt;</code>            | Size of repeating merge container in bits                          |
| <code>--merge-offset &lt;BITS&gt;</code>              | Bit offset of unit inside merge raw unit [default: 0]              |
| <code>--merge-drop-partial-eof</code>                 | Discard incomplete trailing bits at EOF in merge stream            |
| <code>--merge-no-seek</code>                          | Disable seeking specifically on merge file                         |
| <code>--merge-use-seek</code>                         | Explicitly enable seeking on merge file (default: true)            |
| <br>General:                                          |                                                                    |
| <code>-q, --quiet</code>                              | Silence non-fatal warnings and diagnostic summaries                |
| <code>-h, --help</code>                               | Print help                                                         |
| <code>-V, --version</code>                            | Print version                                                      |

## Building, Verifying and Distributing

For developers extending `bdd`, compiling lean embedded binaries, or packaging for Linux distributions, Chapter 14 details the verification test suite, modular feature flags, documentation compiler, and release tooling.

## 14. Development, Testing, Packaging & Releases

Run the test suite:

```
make test
# or: cargo test --all-targets --all-features
```

Run benchmarks:

```
make bench
# or: cargo bench
```

Run code linter and formatting checks:

```
make lint
make check-fmt
```

## Cargo Features & Minimal Footprint Builds

`bdd` supports modular compile-time features in `Cargo.toml`. Both features are enabled by default for maximum out-of-the-box functionality:

| FEATURE                   | ALIASES                                  | DESCRIPTION                                                                                                       | DEFAULT |
|---------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------|
| <code>server</code>       | <code>web</code> ,<br><code>serve</code> | Embedded zero-dependency HTTP/1.1 web application server ( <code>bdd --serve</code> ) and embedded browser assets | Yes     |
| <code>small-floats</code> | <code>small-float</code>                 | Sub-32-bit floating-point codecs (FP16, BF16, FP8 E4M3/E5M2, FP6, FP4) and float presets                          | Yes     |

To compile a lean, minimal-footprint binary without web UI assets or floating-point conversion tables (ideal for embedded environments, containerized microservices, or minimal CI pipelines):

```
# Minimal footprint build (integers, bit manipulation, and stream pipelines only):
cargo build --release --no-default-features

# Build with only small floating-point codecs:
cargo build --release --no-default-features --features small-floats

# Build with only embedded web application server:
cargo build --release --no-default-features --features server
```

## Documentation & PDF Generation

All project documentation compiles into clean, print-ready vector PDF and HTML files:

- [README.pdf](#) : Generated from `README.md` via headless Chromium / Puppeteer with vector math, diagrams, and GFM styling.
- [docs/bdd.1.pdf](#) : Unix manual page rendered as clean vector PDF via `groff` and `ps2pdf`.
- [docs/bdd.1.html](#) : Unix manual page rendered as standalone HTML.
- [docs/Presentation.pdf](#) : Architectural slide deck compiled from OpenDocument Presentation ( `docs/Presentation.odp` ) via headless LibreOffice.

To regenerate all documentation artifacts in one command:

```
make docs
```

## Distribution Packages & Releases

Native packages, Python modules, and C SDK archives can be built directly using the included packaging suite:

- **Debian / Ubuntu / Linux Mint ( `.deb` ):**

```
make deb
# Install: sudo apt install ./dist/bdd-0.5.0-amd64.deb
```

- **Fedora / RHEL / CentOS / Rocky ( `.rpm` for DNF):**

```
make rpm
# Install: sudo dnf install ./dist/bdd-0.5.0-1.*.rpm
```

- **Python Module for Pip ( `.whl` & `.tar.gz` ):**

```
make python
# Install: pip install ./dist/bdd-0.5.0-py3-none-any.whl
```

- **Standalone C Library SDK Archive ( `.tar.gz` ):**

```
make c-lib
# Extract: tar -xzf ./dist/bdd-c-0.5.0-linux-x86_64.tar.gz
```

- **Build All Distribution Packages & Checksums:**

```
make packages
```

All generated distribution packages ( `.deb` , `.rpm` , `.whl` , `.tar.gz` ) are automatically built and published as downloadable assets on [GitHub Releases](#) upon pushing a version tag (e.g. `v0.5.0` ).

---

## License

GPL-3.0-or-later. Original author: Esa Turtiainen.